

Pose Estimation Task Force

Towards foundational models for cross-species models

BBQS Consortium in-person meeting
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Who (please add your email)

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WHO: More TBD (if interested, just email cody.c.baker.phd@gmail.com)

[Ideally everyone doing pose estimation in their daily lives]

GOAL(s):

- Provide universally applicable cross-species foundation models for one-shot pose estimation.
- Shift the norm of pose estimation processing away from local operation towards end-to-end cloud pipelines.
- Identify how to generate and curate enough data for improving robustness of models.

Key Outcomes/Challenges:

- Establish a user-friendly web-only frontend for rapid video frame annotation
- Collect a gold standard meta-collection of high-quality, human-annotated datasets
- Collate and standardize all training data into the common EMBER backend
- Integrate crowd-sourced annotations and establish curation rankings for submissions
- Establish a similarly user-friendly web-only frontend for visualizing the outcomes of model inference and offering quality metric evaluations to ensure high confidence (or feedback mechanism for re-labeling weight)
- Rodent-friendly models versus cross-species

What's next:

- Consolidate as much data from as many labs as possible within, across, and even outside BBQS
- Plan hands-on session at the workshop to introduce this tool to the consortium and educate as many new data annotators as possible